

NATIONAL THERMAL ANOMALY & INDUSTRIAL RISK INTELLIGENCE REPORT

SIH 2026 | NTRO CHALLENGE: PERSISTENT THERMAL SOURCE SURVEILLANCE

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1. Executive Summary & Macro Observations

Total Fire Detections:	1517	Total Thermal Radiative Energy:	22,048.3 MW
Monitored Facilities:	24	Extreme / High Risk Facilities:	2 Extreme, 5 High
Dominant Fire Category:	Forest Fire	Monthly / Quarterly Cycles:	1 Months, 1 Quarters

2. Critical Facility Risk Matrix (Top Priority Assets)

Facility Name	Type	Nearby Fires (<5km)	Peak FRP (MW)	Risk Score	Risk Tier
Talcher Super TPS	thermal_power_pl	45	127.4	82.7 / 100	EXTREME RISK
Talcher Coalfield	mining	13	127.4	81.4 / 100	EXTREME RISK
Korba Coalfield	mining	3	77.3	58.5 / 100	HIGH RISK
SAIL Bhilai Steel Plant	steel_plant	11	9.9	58.3 / 100	HIGH RISK
Korba STPP	thermal_power_pl	2	69.7	56.7 / 100	HIGH RISK
Hazira Gas Processing	gas_flare	7	10.6	52.7 / 100	HIGH RISK
Jamnagar Refinery (Relia	oil_refinery	2	105.2	51.9 / 100	HIGH RISK
Jharia Coalfield	mining	6	3.1	41.7 / 100	MODERATE RISK

3. Monthly Longitudinal Thermal Trend Analysis

Period	Detections	Avg Brightness	Avg FRP	Total Energy (MW)	Near Industry	MoM Velocity
2026-09	1517	56.6 K	14.5 MW	22,048.3	18	Baseline

4. Regional Spatial Surveillance Quadrants

Surveillance Zone	Coverage States	Total Fires	Industrial	Gas Flares	Trend Vector
Northern Agro-Industrial Zone	Punjab, Haryana, UP, NCR	179	0	0	Stable (■)
Western Energy & Petrochemical Corridor	Gujarat, Maharashtra, Offsho	139	8	0	Stable (■)
Central-Eastern Mining & Steel Belt	Odisha, Jharkhand, Chhattisg	712	19	0	Stable (■)
Southern Coastal Infrastructure Zone	Karnataka, Kerala, Tamil Nad	568	0	0	Stable (■)

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Intended for official defense and disaster response operations.